Multilevel Modeling for Longitudinal Data

Zhengxiao Wei, Eleah Stringer

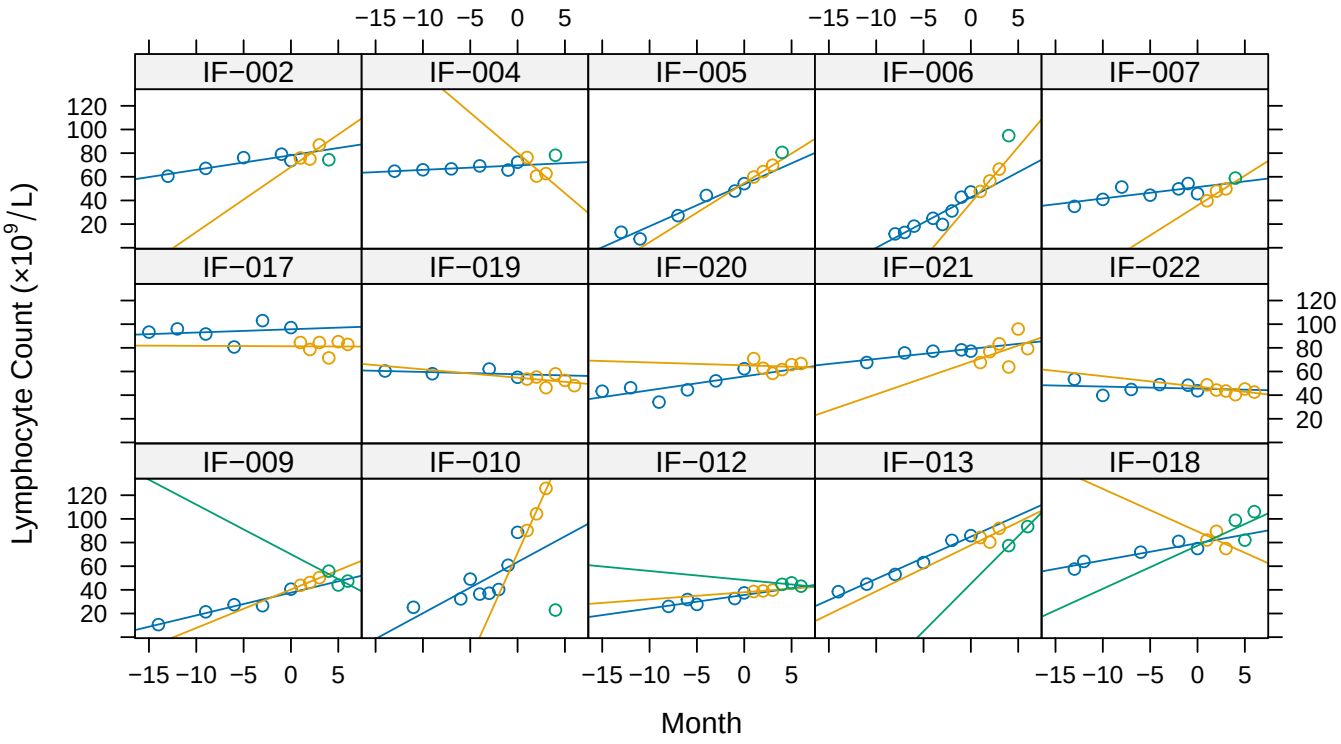
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1. Data description

The longitudinal data are from an experiment to study the effect of intermittent fasting (IF) on the lymphocyte measurements of 15 participants, each of whom was repeatedly measured before, during, and after the intervention. The data set (long format) consists of six columns:

- **LYMP** - [numeric] lymphocyte count ($\times 10^9$/L); lower values usually correspond to better outcomes;
- **ATG** - [numeric] relative net LC3-II expression; higher levels are associated with increased autophagy;
- **ID** - [factor] participant identifier; 15 levels;
- **Time** - [numeric] centered time point, with a frequency of 30 days, treating the **last** pretest as Month 0;
- **Time_Within_Phase** - [numeric] time point within each phase, starting from Month 0;
- **Phase** - [numeric] treatment identifier; 0 as pretest, 1 as intervention, and 2 as posttest.

```
df <- read.csv("~/Downloads/IF_v3.1.csv", stringsAsFactors=T)
df$Phase <- factor(df$Phase)
lattice::xyplot(LYMP ~ Time | ID, data=df, group=Phase, type=c("p","r"), layout=c(5,3),
                index.cond=list(c(6:9,11,10,12:15,1:5)), #change panel order
                xlab="Month", ylab=expression(paste("Lymphocyte Count (\u00d7",10^9/L,")")))
```



2. Linear mixed-effects regression (Part I)

The hypothesis is that IF can help patients lower their lymphocyte count. The objectives are then to investigate whether (1) the mean responses and (2) the linear slopes of time are statistically different before versus during the intervention.

2.1 Using the "*nlme*" package (v 3.1-163)

```
dat <- df[!is.na(df$LYMP) & df$Phase!=2,] #IF-013 contains one missing value

fit1 <- nlme::lme(fixed=LYMP ~ Phase,
                  random=~1 | ID,         #random intercept for each subject
                  data=dat, method="ML")
summary(fit1)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: dat
##        AIC      BIC    logLik
##   1237.862 1249.851 -614.9311
##
## Random effects:
##  Formula: ~1 | ID
##         (Intercept) Residual
## StdDev:    15.93324 13.46224
##
## Fixed effects:  LYMP ~ Phase
##                Value Std.Error  DF  t-value p-value
## (Intercept) 52.52277  4.391316 132 11.96060       0
## Phase1      11.90878  2.318019 132  5.13748       0
##  Correlation:
##        (Intr)
## Phase1 -0.211
##
## Standardized Within-Group Residuals:
##         Min          Q1         Med          Q3         Max
## -2.51108775 -0.56042850  0.06447024  0.60540456  4.07260524
##
## Number of Observations: 148
## Number of Groups: 15
```

`~1 | ID` is the random effects part of the model, allowing for the possibility that different subjects have different baseline lymphocyte counts. `LYMP ~ Phase` specifies the fixed effect (as the *aov* and *lm* functions usually do), expecting the dependent variable to be influenced by the categorical treatment identifier.

This tutorial model (`fit1`) compares the dependent variable (`LYMP`) between the different phases of the experiment (`Phase`, dummy variable), while controlling for the repeated measures within each subject.

Two methods are available to estimate the coefficients: the restricted maximum likelihood (REML) by default and the maximum likelihood (ML). In a nutshell, ML is generally preferred when comparing models with different fixed effects.

```r
fit2 <- nlme::lme(fixed=LYMP ~ Phase * Time,
                  random=~Time | ID,      #random intercept and slope for each subject
                  data=dat, method="ML")
summary(fit2)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: dat
##        AIC      BIC    logLik
##   1126.297 1150.274 -555.1483
##
## Random effects:
##  Formula: ~Time | ID
##  Structure: General positive-definite, Log-Cholesky parametrization
##             StdDev    Corr
## (Intercept) 16.171106 (Intr)
## Time         2.047608 0.05
## Residual     7.208301
##
## Fixed effects:  LYMP ~ Phase * Time
##                 Value Std.Error  DF   t-value p-value
## (Intercept) 61.57578  4.421432 130 13.926659  0.0000
## Phase1       1.04890  2.406490 130  0.435863  0.6637
## Time         1.67659  0.565761 130  2.963427  0.0036
## Phase1:Time -0.05164  0.737775 130 -0.069994  0.9443
##  Correlation:
##             (Intr) Phase1 Time
## Phase1      -0.153
## Time         0.117 -0.136
## Phase1:Time -0.056 -0.615 -0.072
##
## Standardized Within-Group Residuals:
##          Min           Q1          Med           Q3          Max
## -3.385032180 -0.367787634 -0.008040879  0.423968276  3.672631587
##
## Number of Observations: 148
## Number of Groups: 15
```

`~Time | ID` adds random slopes for the time variable, which will allow the effect of time to vary across subjects. The interaction term `Phase * Time` allows the slope of the time trend to differ between the phases.

```r
(coef.fixed <- coef(summary(fit2))[,"Value"]) #extract fixed effect coefficients
```

```
## (Intercept)      Phase1        Time Phase1:Time
## 61.57578172  1.04890005  1.67659222 -0.05163969
```

```r
fit2u <- nlme::lme(fixed=LYMP ~ Phase * Time, random=~Phase * Time | ID,
                   data=dat, method="ML",             #maximal random effects
                   control=nlme::lmeControl(opt="optim", optCtrl=list(maxit=1e5)))
                                          #allows more iterations for convergence
fit3u <- nlme::lme(fixed=LYMP ~ Phase + Time, random=~Phase + Time | ID,
                   data=dat, method="ML",
                   control=nlme::lmeControl(opt="optim", optCtrl=list(maxit=1e5)))
```

Furthermore, `~Phase * Time | ID` specifies a model (`fit2u`) with an interaction term as a random effect. This specification would allow the intercept, the slope for time, the slope for phase, and the slope for the interaction between time and phase to vary across subjects. However, please note that this model is quite complex and may require a large amount of data to estimate reliably. The `boundary (singular) fit` warning indicates that the estimated variance-covariance matrix is non-positive definite.

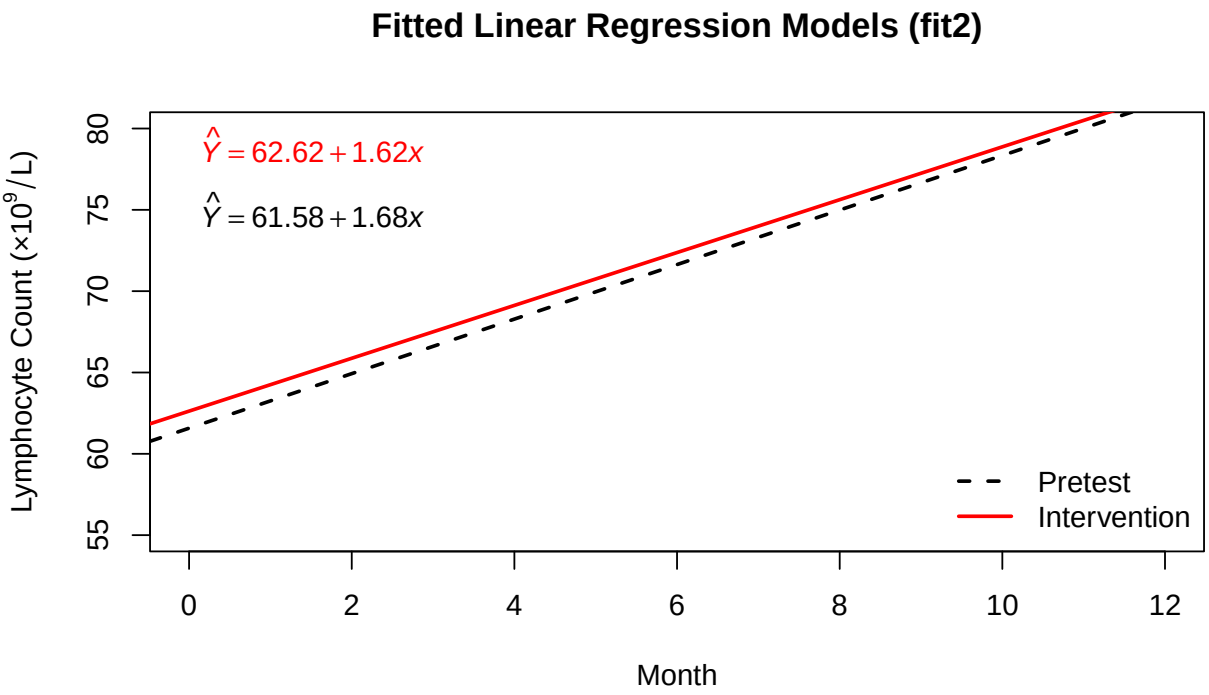
```
# coef(fit2u) #all coefficients vary across subjects
coef(fit2)    #the intercept and the slope for time vary across subjects
```

```
##        (Intercept) Phase1       Time Phase1:Time
## IF-002    76.64484 1.0489  1.12003288 -0.05163969
## IF-004    66.97597 1.0489  0.09173338 -0.05163969
## IF-005    55.16979 1.0489  3.66212044 -0.05163969
## IF-006    44.93761 1.0489  4.57638488 -0.05163969
## IF-007    47.77781 1.0489  0.57825736 -0.05163969
## IF-009    40.76422 1.0489  2.22585463 -0.05163969
## IF-010    78.17523 1.0489  6.75245040 -0.05163969
## IF-012    36.59460 1.0489  1.25931181 -0.05163969
## IF-013    80.04303 1.0489  3.03665220 -0.05163969
## IF-017    85.18630 1.0489 -0.69445776 -0.05163969
## IF-018    78.18521 1.0489  1.32756347 -0.05163969
## IF-019    54.29697 1.0489 -0.50018280 -0.05163969
## IF-020    58.04264 1.0489  1.35134812 -0.05163969
## IF-021    75.98748 1.0489  0.61256766 -0.05163969
## IF-022    44.85502 1.0489 -0.25075341 -0.05163969
```

The function *coef* is extracting individual fitted intercept and slope in each phase. For example, IF-002 has a lymphocyte baseline of 76.64484 and a growth rate of 1.12003288 before the intervention (`Phase==0`); and a lymphocyte baseline of $76.64484+1.0489 = 77.69374$ and a growth rate of $1.12003288-0.05163969 = 1.06839319$ during the intervention (`Phase==1`). IF-017, 019, and 022 have a negative growth rate.

These values are a combination of the fixed and random effects. Use `coef` and `summary` to obtain the coefficients of the fixed effects, which are the mean fitted intercept and slope of all 15 subjects. On Page 3, the mean lymphocyte baseline is 61.57578172, and the mean growth rate is 1.67659222 before the intervention. The mean lymphocyte baseline becomes $61.57578172 + 1.04890005 = 62.62468177$, and the mean growth rate becomes $1.67659222 - 0.05163969 = 1.62495253$ during the intervention.

```
par(mar=c(5,5,5,5))
plot(1, xlim=c(0, 12), ylim=c(55, 80), main="Fitted Linear Regression Models (fit2)",
     type="n", xlab="Month", ylab=expression(paste("Lymphocyte Count (\u00d7",10^9/L,")")))
abline(a=coef.fixed[1], b=coef.fixed[3], lty=2, lwd=2)
abline(a=coef.fixed[1]+coef.fixed[2], b=coef.fixed[3]+coef.fixed[4], col="red", lwd=2)
legend("bottomright", c("Pretest", "Intervention"),
       col=c("black","red"), lty=c(2,1), lwd=2, bty="n")
text(1.5,75,bquote(hat(italic(Y)) ==
                   .(format(coef.fixed[1],digits=4,nsmall=2))
                   + .(round(coef.fixed[3],2)) * italic(x)))
text(1.5,79,bquote(hat(italic(Y)) == .(round(coef.fixed[1]+coef.fixed[2],2))
                 + .(round(coef.fixed[3]+coef.fixed[4],2)) * italic(x)), col="red")
```



The 95% confidence intervals for the fixed effects.

```
nlme::intervals(fit2)
```

```
## Approximate 95% confidence intervals
##
##  Fixed effects:
##                 lower        est.     upper
## (Intercept) 52.9475227 61.57578172 70.204041
## Phase1      -3.6472747  1.04890005  5.745075
## Time         0.5725303  1.67659222  2.780654
## Phase1:Time -1.4913799 -0.05163969  1.388101
##
##  Random Effects:
##   Level: ID
##                             lower        est.      upper
## sd((Intercept))       11.1996358 16.17110565 23.3493894
## sd(Time)               1.3994807  2.04760782  2.9958953
## cor((Intercept),Time) -0.4281287  0.04973367  0.5058618
##
##  Within-group standard error:
##    lower     est.    upper
## 6.343500 7.208301 8.191000
```

2.2 Using the "*lme4*" package (v 1.1-34)

```
fit1.alt <- lme4::lmer(LYMP ~ Phase + (1 | ID), dat, REML=FALSE) #(same) output omitted

fit2.alt <- lme4::lmer(LYMP ~ Phase * Time + (Time | ID), dat, REML=FALSE)

fit3.alt <- lme4::lmer(LYMP ~ Phase + Time + (Time | ID), dat, REML=FALSE)

fit2u.alt <- lme4::lmer(LYMP ~ Phase * Time + (Phase * Time | ID), dat, REML=F,
                        control=lme4::lmerControl(optCtrl=list(maxfun=1e5)))

fit3u.alt <- lme4::lmer(LYMP ~ Phase + Time + (Phase + Time | ID), dat, REML=F,
                        control=lme4::lmerControl(optCtrl=list(maxfun=1e5)))
```

```
#95% prediction intervals for each subject at their data points
head(cbind(dat, merTools::predictInterval(fit2.alt)), 9)
```

```
##     LYMP  ATG     ID Time Time_Within_Phase Phase      fit      upr      lwr
## 1  60.50   NA IF-002  -13                 0     0 62.50886 78.11544 46.76991
## 2  67.10   NA IF-002   -9                 4     0 67.40679 80.47250 54.21268
## 3  76.20   NA IF-002   -5                 8     0 71.24944 83.78322 59.96324
## 4  79.10   NA IF-002   -1                12     0 75.24281 87.48445 64.15489
## 5  73.70 0.31 IF-002    0                13     0 76.44063 87.37349 64.88781
## 6  75.90 0.20 IF-002    1                 0     1 78.72011 90.20285 67.05512
## 7  75.00   NA IF-002    2                 1     1 79.98366 91.89481 67.57593
## 8  86.90 0.22 IF-002    3                 2     1 80.20423 92.55541 68.17402
## 10 64.76   NA IF-004  -13                 0     0 66.47387 82.35032 50.27933
```

The "*lme4*" package does not provide standard errors for the random effects by default. However, the "*merTools*" package offers functions to help with constructing prediction intervals.

2.3 Using the "*BayesFactor*" package (v 0.9.12-4.5)

```
set.seed(277)                       #cf. fit2
full <-     BayesFactor::lmBF(LYMP~Time*Phase+Time:ID+ID, dat, whichRandom="ID", progress=F)

set.seed(277)                       #cf. fit3
additive <- BayesFactor::lmBF(LYMP~Time+Phase+Time:ID+ID, dat, whichRandom="ID", progress=F)

additive / full #anecdotal evidence of no interaction effect
```

```
## Bayes factor analysis
## --------------
## [1] Time + Phase + Time:ID + ID : 2.688412 ±5.98%
##
## Against denominator:
##   LYMP ~ Time * Phase + Time:ID + ID
## ---
## Bayes factor type: BFlinearModel, JZS
```

See also Bub, Masson, and van Noordenne (2021, p. 80), van den Bergh et al. (2023), and Wei, Nathoo, and Masson (2023).

2.4 Summary

In a linear mixed-effects regression (LMER), each subject has its intercept and slope (i.e., random intercepts and random slopes).

The p-value of .9443 for the interaction term in `fit2` is not statistically significant at an alpha level of .05 ($t = -0.069994$, $df = 130$). Hence, fail to reject that the linear slopes of time are the same before versus during the intervention.

▶ In summary, the mean lymphocyte count continued to increase at the same rate after participants received IF treatment.

If the slope of the time trend is the same across phases, one can remove the interaction term and specify `Phase + Time` as fixed effects (i.e., an additive model).

```
fit3 <- nlme::lme(fixed=LYMP ~ Phase + Time, #additive model
                  random=~Time | ID,
                  data=dat, method="ML")
summary(fit3)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: dat
##        AIC      BIC    logLik
##   1124.302 1145.282 -555.1508
##
## Random effects:
##  Formula: ~Time | ID
##  Structure: General positive-definite, Log-Cholesky parametrization
##             StdDev    Corr
## (Intercept) 16.169286 (Intr)
## Time         2.049480 0.05
## Residual     7.207798
##
## Fixed effects:  LYMP ~ Phase + Time
##                Value Std.Error  DF   t-value p-value
## (Intercept) 61.55864  4.398764 131 13.994531  0.0000
## Phase1       0.94518  1.890170 131  0.500051  0.6179
## Time         1.67379  0.562812 131  2.973985  0.0035
##  Correlation:
##        (Intr) Phase1
## Phase1 -0.238
## Time    0.114 -0.229
##
## Standardized Within-Group Residuals:
##          Min           Q1          Med           Q3          Max
## -3.385301421 -0.368378691 -0.009949635  0.428584644  3.665722929
##
## Number of Observations: 148
## Number of Groups: 15
```

3. Linear mixed-effects regression (Part II)

Investigate any difference in either (3) autophagy or (4) lymphocyte count between those who fasted for three versus six months.

```
lymphocyte <- df[df$Phase==1, -2]
autophagy <- df[!is.na(df$ATG) & df$Phase==1, -1] #Excluded IF-007 and 012 (no records)
fast6 <- paste0("IF-0",c(17,19,20,21,22))
lymphocyte$Fast <- factor(ifelse(lymphocyte$ID %in% fast6, "Six", "Three"))
autophagy$Fast <- factor(ifelse(autophagy$ID %in% fast6, "Six", "Three"))

fit.LYMP <- nlme::lme(fixed=LYMP ~ Fast * Time_Within_Phase,
                      random=~Time_Within_Phase | ID, data=lymphocyte, method="ML",
                      control=nlme::lmeControl(opt="optim", optCtrl=list(maxit=1e5)))
summary(fit.LYMP)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: lymphocyte
##        AIC      BIC    logLik
##   462.5947 479.3495 -223.2974
##
## Random effects:
##  Formula: ~Time_Within_Phase | ID
##  Structure: General positive-definite, Log-Cholesky parametrization
##                    StdDev    Corr
## (Intercept)       16.118636 (Intr)
## Time_Within_Phase  1.524995 0.976
## Residual           6.486373
##
## Fixed effects:  LYMP ~ Fast * Time_Within_Phase
##                                Value Std.Error DF   t-value p-value
## (Intercept)                 63.50857  7.771496 43  8.171988  0.0000
## FastThree                  -0.17024  9.591116 13 -0.017750  0.9861
## Time_Within_Phase           0.17657  1.006740 43  0.175389  0.8616
## FastThree:Time_Within_Phase  3.83843  1.875263 43  2.046875  0.0468
##  Correlation:
##                            (Intr) FstThr Tm_W_P
## FastThree                  -0.810
## Time_Within_Phase           0.492 -0.399
## FastThree:Time_Within_Phase -0.264  0.232 -0.537
##
## Standardized Within-Group Residuals:
##         Min          Q1         Med          Q3         Max
## -2.26851249 -0.42347096 -0.03517308  0.34098717  2.48387850
##
## Number of Observations: 60
## Number of Groups: 15
```

```r
fit.ATG <- nlme::lme(fixed=ATG ~ Fast * Time_Within_Phase,
                     random=~Time_Within_Phase | ID,
                     data=autophagy, method="ML")
summary(fit.ATG)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: autophagy
##       AIC      BIC   logLik
##   1.33751 16.47207 7.331245
##
## Random effects:
##  Formula: ~Time_Within_Phase | ID
##  Structure: General positive-definite, Log-Cholesky parametrization
##                    StdDev     Corr
## (Intercept)       0.28833162 (Intr)
## Time_Within_Phase 0.08691071 -0.902
## Residual          0.14765105
##
## Fixed effects:  ATG ~ Fast * Time_Within_Phase
##                                    Value  Std.Error DF   t-value p-value
## (Intercept)                   0.5887741 0.14723033 34  3.999000  0.0003
## FastThree                     0.0182063 0.18847409 11  0.096598  0.9248
## Time_Within_Phase            -0.0307843 0.04514058 34 -0.681966  0.4999
## FastThree:Time_Within_Phase -0.0885907 0.06744962 34 -1.313434  0.1978
##  Correlation:
##                              (Intr) FstThr Tm_W_P
## FastThree                    -0.781
## Time_Within_Phase            -0.893  0.697
## FastThree:Time_Within_Phase   0.597 -0.825 -0.669
##
## Standardized Within-Group Residuals:
##        Min         Q1        Med         Q3        Max
## -1.4199410 -0.6592750 -0.1162571  0.4896875  1.9368328
##
## Number of Observations: 49
## Number of Groups: 13
```

▶ The p-value of .0468 for the interaction term in `fit.LYMP` is statistically significant at an alpha level of .05 ($t = 2.046875$, $df = 43$). Hence, reject that the linear slopes of time are the same between the two fasting groups. Participants who fasted for three months showed a higher lymphocyte growth rate than those who fasted for six months during the intervention.

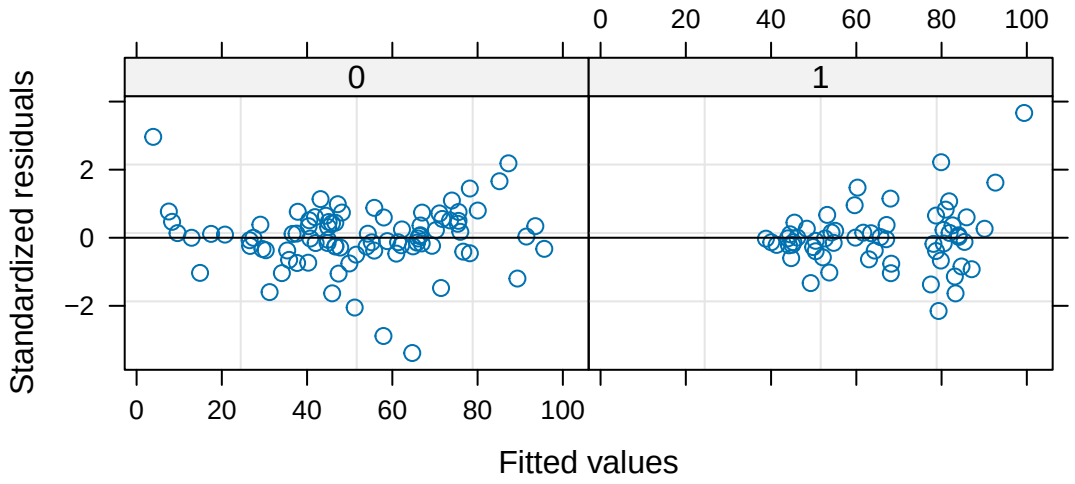
▶ The p-value of .1978 for the interaction term in `fit.ATG` is not statistically significant at an alpha level of .05 ($t = -1.313434$, $df = 34$). Hence, fail to reject that the linear slopes of time are the same between the two fasting groups. There was no difference between the autophagy values of fasting for three and six months.

4. Diagnostics

4.1 Check for linearity and homoscedasticity

Plot standardized residuals versus fitted values by phase. One should see a random scattering of points if the assumptions are met. Plus, check for influential outliers.

```
plot(fit3, resid(., type = "p") ~ fitted(.) | Phase, abline=0)
```



The intervention phase (right panel, `Phase==1`) may witness heteroscedasticity according to the residual plot. Using the `weights` argument, one can incorporate a variance function structure into the model. The variance structure `varIdent(form=~1 | Phase)` allows for different variances for each phase level.

```
fit4 <- nlme::lme(fixed=LYMP ~ Phase + Time,
                  random=~Time | ID, data=dat, method="ML",
                  cor=nlme::corCompSymm(),
                  weights=nlme::varIdent(form=~1 | Phase),
                  control=nlme::lmeControl(opt="optim", optCtrl=list(maxit=1e5)))
summary(fit4)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: dat
##        AIC      BIC    logLik
##   1128.256 1155.231 -555.1282
##
## Random effects:
##  Formula: ~Time | ID
##  Structure: General positive-definite, Log-Cholesky parametrization
##             StdDev    Corr
## (Intercept) 16.178759 (Intr)
## Time         2.068281 0.059
## Residual     7.321290
##
## Correlation Structure: Compound symmetry
##  Formula: ~1 | ID
```

```
##  Parameter estimate(s):
##         Rho
## 0.001085912
## Variance function:
##  Structure: Different standard deviations per stratum
##  Formula: ~1 | Phase
##  Parameter estimates:
##         0         1
## 1.0000000 0.9610993
## Fixed effects:  LYMP ~ Phase + Time
##                Value Std.Error  DF   t-value p-value
## (Intercept) 61.55609  4.406474 131 13.969465  0.0000
## Phase1       0.96305  1.895686 131  0.508022  0.6123
## Time         1.67643  0.568174 131  2.950560  0.0038
##  Correlation:
##        (Intr) Phase1
## Phase1 -0.243
## Time    0.123 -0.232
##
## Standardized Within-Group Residuals:
##          Min           Q1          Med           Q3          Max
## -3.376270259 -0.364140887 -0.008806914  0.426628645  3.645489602
##
## Number of Observations: 148
## Number of Groups: 15
```

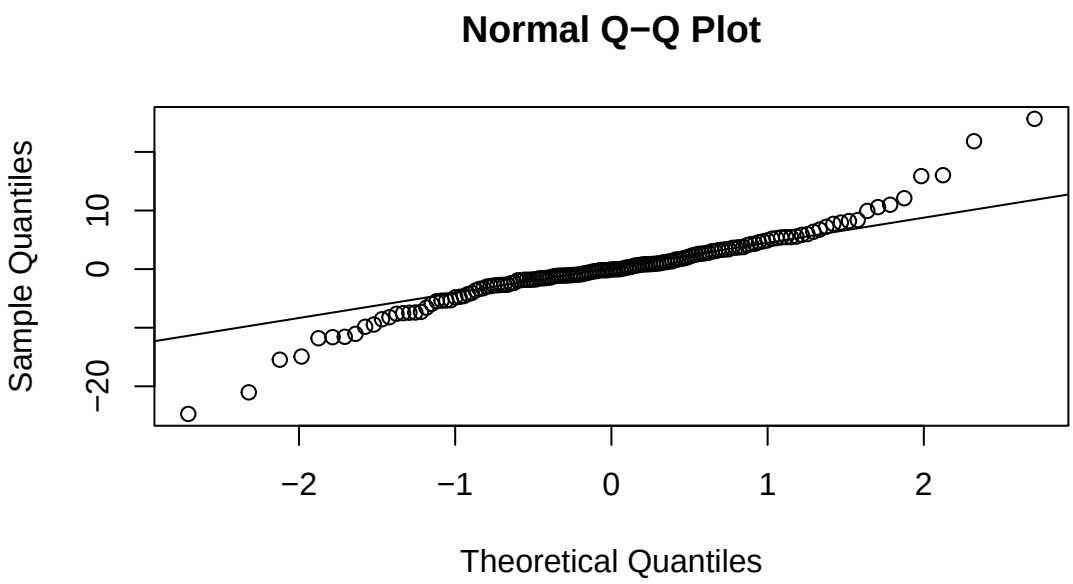
```
coef(fit4) #final model
```

```
##        (Intercept)    Phase1        Time
## IF-002    76.60368 0.9630503  1.11919252
## IF-004    66.88368 0.9630503  0.07954691
## IF-005    55.21804 0.9630503  3.66630985
## IF-006    45.01679 0.9630503  4.58925502
## IF-007    47.63299 0.9630503  0.56191905
## IF-009    40.80104 0.9630503  2.22770109
## IF-010    78.66740 0.9630503  6.83937153
## IF-012    36.56710 0.9630503  1.24902880
## IF-013    79.95425 0.9630503  3.02854535
## IF-017    84.99424 0.9630503 -0.71052974
## IF-018    78.16515 0.9630503  1.32507180
## IF-019    54.18402 0.9630503 -0.51410675
## IF-020    58.00937 0.9630503  1.34323753
## IF-021    75.87096 0.9630503  0.60685324
## IF-022    44.77264 0.9630503 -0.26492011
```

4.2 Check for normality of the residuals

The residuals should follow a straight line in a QQ plot (Quantile-Quantile plot) if normally distributed.

```
qqnorm(resid(fit4)); qqline(resid(fit4)) # shapiro.test(resid(fit4))
```



4.3 Check for random effects assumptions

Check that the random effects are zero-centered and normally distributed, and have a constant variance. Use `ranef` to extract the random effects from the model and plot histograms.

```
rand.ef <- nlme::ranef(fit4)$Time #15 subjects
mean(rand.ef)
```

```
## [1] -1.480297e-16
```

```
hist(rand.ef, prob=TRUE, col="white", yaxt="n",
     ylab="", xlab="Subject-Specific Random Effects (15)", main="")
lines(density(rand.ef),col="red",lwd=2)
```

